BSDS V: Regression Techniques

Problems Set 3: Departure from Assumptions of Normal Linear Model

Throughout this problem set, we will consider the multiple linear regression model with response variable y and a list of regressors/covariates $\{x_1, \ldots, x_p\}$. Based on n observations the model is

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad \text{where} \quad \underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} \underset{\sim}{1} & \underset{\sim}{x}_{(1)} & \ldots & \underset{\sim}{x}_{(p)} \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underset{\sim}{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}. \tag{1}$$

We assume that $E(\underset{\sim}{\varepsilon}) = \underset{\sim}{0}$ and $var(\underset{\sim}{\varepsilon}) = \sigma^2 I_n$.

1. Let the underlying actual model be (1). However, you have collected data on q additional variables $\{z_1, \ldots, z_q\}$ and work on the model:

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + Z\underset{\sim}{\gamma} + \underset{\sim}{\epsilon}^\star, \quad \text{where } \underset{\sim}{\epsilon}^\star \sim N(\underset{\sim}{0}, \sigma^2 I), \quad Z = \begin{bmatrix} \underset{\sim}{z}_{(1)} & \ldots & \underset{\sim}{z}_{(q)} \end{bmatrix} \tag{2}$$

   Consider a new observation $(\underset{\sim}{x}_0^\top : \underset{\sim}{z}_0^\top)$ with corresponding response Y_0. Compare the confidence interval of $\underset{\sim}{x}_0^\top\underset{\sim}{\beta}$ obtained from your model (4)with the one obtained from the actual model (1). Also, compare the prediction interval for Y_0 obtained from model (4) with the one obtained from the actual model (1).

2. Consider the linear model in (1) as your working model. Suppose, however, that the underlying true model is as in (1) except for the fact that $\text{var}(\underset{\sim}{\epsilon}) = \sigma^2\text{Diag}(t, t^2, \ldots, t^n)$ for some number $t > 1$. How would this misspecification affect the variance of OLS $\widehat{\underset{\sim}{\beta}}_{\text{OLS}}$ and expectation of $\widehat{\sigma}^2_{\text{UE}} = \text{RSS}/(n-p-1)$?

3. Consider the above setup in Q.2. However, suppose that you know t. How would you modify the regression estimate of $\underset{\sim}{\beta}$ so that its variance is smaller than the variance of $\widehat{\underset{\sim}{\beta}}_{\text{OLS}}$ obtained in Q.2 (in the sense that the difference of the two covariance matrices is negative semi-definite).

4. Let $\underset{\sim}{x}_{(1)}, \ldots, \underset{\sim}{x}_{(n)}$ be n observations on p-covariates, i.e., $\underset{\sim}{x}_{(i)} \in \mathbb{R}^p$, and define $\bar{\underset{\sim}{x}} = n^{-1}\sum_{i=1}^n \underset{\sim}{x}_{(i)}$, and $S_X = n^{-1}(\underset{\sim}{x}_{(i)} - \bar{\underset{\sim}{x}})(\underset{\sim}{x}_{(i)} - \bar{\underset{\sim}{x}})^\top$. Consider the Mahalanobis distances:

MD_i	$\left\{(\underset{\sim}{x}_{(i)} - \bar{\underset{\sim}{x}})^\top S_X^{-1}(\underset{\sim}{x}_{(i)} - \bar{\underset{\sim}{x}})\right\}^{1/2}$	MD of $\underset{\sim}{x}_{(i)}$ from the center,
$\text{MD}_{i,j}$	$\left\{(\underset{\sim}{x}_{(i)} - \underset{\sim}{x}_{(j)})^\top S_X^{-1}(\underset{\sim}{x}_{(i)} - \underset{\sim}{x}_{(j)})\right\}^{1/2}$	MD between $\underset{\sim}{x}_{(i)}$ and $\underset{\sim}{x}_{(j)}$.

   Suppose $\underset{\sim}{x}_{(i)}$s are i.i.d. samples from a p-variate distribution with finite variance-covariance matrix.

   (a) Show that $E(\text{MD}_i^2) = p$.

   (b) Show that $E(\text{MD}_{i,j}^2) = 2np/(n-1)$.

   (c) Define $Z = \begin{bmatrix} \underset{\sim}{x}_{(1)}^\top \\ \vdots \\ \underset{\sim}{x}_{(n)}^\top \end{bmatrix}$, $X = \begin{bmatrix} \underset{\sim}{1} & Z \end{bmatrix}$ and $P(X) = X(X'X)^{-1}X' = ((p_{i,j}))$. Using (i) and (ii), derive the expectations of $p_{i,i}$ and $p_{i,j}$ for $i \neq j$.

5. Let $\underset{\sim}{Y}$ be the vector of n-observations on a response variable Y, $\{\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_p, \underset{\sim}{z}_1, \ldots, \underset{\sim}{z}_q\}$ be the corresponding vectors for $p+q$ covariates, and $X = \begin{bmatrix} \underset{\sim}{1} & \underset{\sim}{x}_1 & \ldots & \underset{\sim}{x}_p \end{bmatrix}$, $Z = \begin{bmatrix} \underset{\sim}{z}_1 & \ldots & \underset{\sim}{z}_q \end{bmatrix}$ and $W = [X : Z]$. Consider two models:

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + Z\underset{\sim}{\gamma} + \underset{\sim}{\epsilon}^\star = W\underset{\sim}{\delta} + \underset{\sim}{\epsilon}^\star; \tag{3}$$

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\epsilon}, \tag{4}$$

where $E(\underset{\sim}{\epsilon}) = E(\underset{\sim}{\epsilon}^\star) = \underset{\sim}{0}$, and $\text{var}(\underset{\sim}{\epsilon}) = \text{var}(\underset{\sim}{\epsilon}^\star) = \sigma^2 I$. Let $\widehat{\underset{\sim}{\gamma}}_{\text{OLS}}$ be the OLS estimator of $\underset{\sim}{\gamma}$ based on model (3). Then show that

$$\underset{\sim}{Y}^\top (P(X) - P(W))\underset{\sim}{Y} = \sigma^2 \widehat{\underset{\sim}{\gamma}}_{\text{OLS}}^\top \left[\text{var}(\widehat{\underset{\sim}{\gamma}}_{\text{OLS}})\right]^{-1} \widehat{\underset{\sim}{\gamma}}_{\text{OLS}}$$

6. Consider fitting a full rank multiple regression model $\underset{\sim}{y} = X\underset{\sim}{\beta} + \underset{\sim}{\epsilon}$ by the least squares method. Let $P(X) = ((p_{ij}))$ be the orthogonal projection matrix onto $\mathcal{C}(X)$ and $P_{(i)} = ((p_{jk(i)}))$ be the same when the ith observation was removed. Let $\underset{\sim}{x}_i$ be the ith row of X (as a column vector), e_i be the ith residual from the full model fit, and $\hat{\underset{\sim}{\beta}}_{(i)}$ be the ith deleted OLS estimate of $\underset{\sim}{\beta}$.

   (a) Show that $e_{i(i)} = \dfrac{e_i}{1 - p_{ii}}$, where $e_{i(i)} = y_i - \underset{\sim}{x}_i^T \hat{\underset{\sim}{\beta}}_{(i)}$.

   (b) Prove that
   $$p_{jk(i)} = p_{jk} + \frac{p_{ik}p_{ij}}{1 - p_{ii}}. \qquad [\text{You may assume } \max\{j,k\} < i]$$

   (c) In case of no intercept model, show that $-0.5 \le p_{ij} \le 0.5, \quad i \ne j$.

7. Consider fitting a regression model with regression matrix X, response vector $\underset{\sim}{Y}$ and unknown parameter vector $\underset{\sim}{\beta}$. Let $\hat{\underset{\sim}{\beta}}$ be the OLS estimate of $\underset{\sim}{\beta}$.

   Suppose that a new observation $(\underset{\sim}{x}_{n+1}, Y_{n+1})$ is added. Show that the residual sum of squares is increased by an amount $\dfrac{e_{n+1}^2}{1 + \underset{\sim}{x}'_{n+1}(X'X)^{-1}\underset{\sim}{x}_{n+1}}$, where $e_{n+1} = Y_{n+1} - \underset{\sim}{x}'_{n+1}\hat{\underset{\sim}{\beta}}$.

8. Consider the model in (1), except that $\underset{\sim}{\varepsilon} \sim N(\underset{\sim}{0}, \sigma^2 V)$ where $V = (1-\rho)I + \rho \underset{\sim}{1}\underset{\sim}{1}^\top$, for some $\rho > 0$. Find an unbiased estimator of σ^2 when ρ is known. Express the unbiased estimator in terms on RSS, $(\bar{Y}, \bar{x}_1, \ldots, \bar{x}_p)$, (n, p) and ρ.

9. In a linear regression setup, let Y be the response and X be a predictor of Y. Based on n observations, consider the model

$$Y_t = \alpha + \beta X_t + \epsilon_t, \text{ where } \epsilon_1 \sim N\left(0, \frac{\sigma^2}{1-\rho^2}\right), \ \epsilon_t = \rho\epsilon_{t-1} + u_t, \ t = 2, \ldots, T, \ \underset{\sim}{u} = \begin{bmatrix} u_2 \\ \vdots \\ u_T \end{bmatrix} \sim N_{T-1}\left(\underset{\sim}{0}, \sigma^2 I\right),$$

   and $\epsilon_1 \perp\!\!\!\perp \underset{\sim}{u}$. Find the variance-covariance matrix and hence the distribution of $\underset{\sim}{\epsilon}$.

10. Consider the model in (1). Recall the variance inflation factor VIF_j, $j = 1, \ldots, p$, and the condition number $\kappa(X)$. Suppose, the covariates are scaled by their receptive standard deviations. Will the value of VIF_j change after this scaling? Will the value of $\kappa(X)$ change after this scaling? Justify your answers.

11. Consider the regression model in (1), and let $P = P(X) = X(X'X)^{-1}X'$ be the hat matrix.

(a) Express ϵ_i in the form $e_i = \underset{\sim}{c}_i'(I_n - P)\underset{\sim}{\epsilon}$ for a suitable vector $\underset{\sim}{c}_i$.

(b) Show that $(n-p-1)^{-1}r_i^2$ can be written as

$$(n-p-1)^{-1}r_i^2 = \frac{Z'QZ}{Z'(I_n - P)Z'}$$

where

$$Q = (1-p_{ii})^{-1}(I_n - P)\underset{\sim}{c}_i\underset{\sim}{c}_i'(I_n - P)$$

r_i is the standardized residual, and $\underset{\sim}{Z} \sim N_n(\underset{\sim}{0}, I_n)$.

(c) Prove that Q is a projection matrix (i.e., show that $Q^2 = Q$ and $Q' = Q$).

(d) Show that $I_n - P - Q$ is a projection matrix and prove that $\underset{\sim}{Z}'Q\underset{\sim}{Z}$ and $\underset{\sim}{Z}'(I_n - P - Q)\underset{\sim}{Z}$ are independent. Hence prove that $(n-p-1)^{-1}r_i^2$ has a Beta $\left(\frac{1}{2}, \frac{1}{2}(n-p)-1\right)$ distribution.

12. Let r_i, $i = 1, \ldots, n$, be the standardized residuals.

(a) Show that,

$$r_i \overset{D}{=} \frac{T\sqrt{n-p-1}}{\sqrt{T^2 + n - p - 2}}, \qquad \text{where} \qquad T \sim t_{n-p-2}.$$

(b) Show that the conditional distribution of T, given $T^2 = s$, is a two-point distribution with equal probability mass at $\pm\sqrt{s}$.

(c) Hence, or otherwise, show that $E(r_i) = 0$.

(d) Show that $\text{var}(r_i) = 1$.

13. Consider the normal linear regression model in (1). Suppose the Gram matrix $X^\top X$ has an eigenvalue 0.001 with corresponding eigenvector $\underset{\sim}{u} = \begin{bmatrix}1, & -1, & 0, & \ldots, & 0\end{bmatrix}^\top$.

Suppose that, due to a compilation error, the response of the last observation is perturbed by 0.1, while all predictor values remain unchanged. Thus, the perturbed response vector and design matrix are given by

$$\underset{\sim}{Y}_p = \underset{\sim}{Y} + \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0.1 \end{bmatrix}, \qquad X_p = X.$$

Given that $x_{n,1} = 0.5$, determine the difference between the estimated contrast $\hat{\beta}_0 - \hat{\beta}_1$ obtained using the perturbed data and that obtained using the original data. Explain how the small eigenvalue of $X^\top X$ influences the sensitivity of this contrast to the perturbation.

14. Let $(\underset{\sim}{Y}_{(-D)}, X_{(-D)})$ be a subset of $(\underset{\sim}{Y}, X)$ of after deleting a set D of d-many observations. Show that the ratio of determinants of the variances of $\widehat{\underset{\sim}{\beta}}$ and $\widehat{\underset{\sim(-D)}{\beta}}$ can be expressed as $\det^{-1}(I - P(D))$,where $P(D)$ is the sub-matrix of $P(X)$ considering the indices corresponding to D.

($\star$) Rectified. *Thanks to Antas for pointing out the anomalies.*